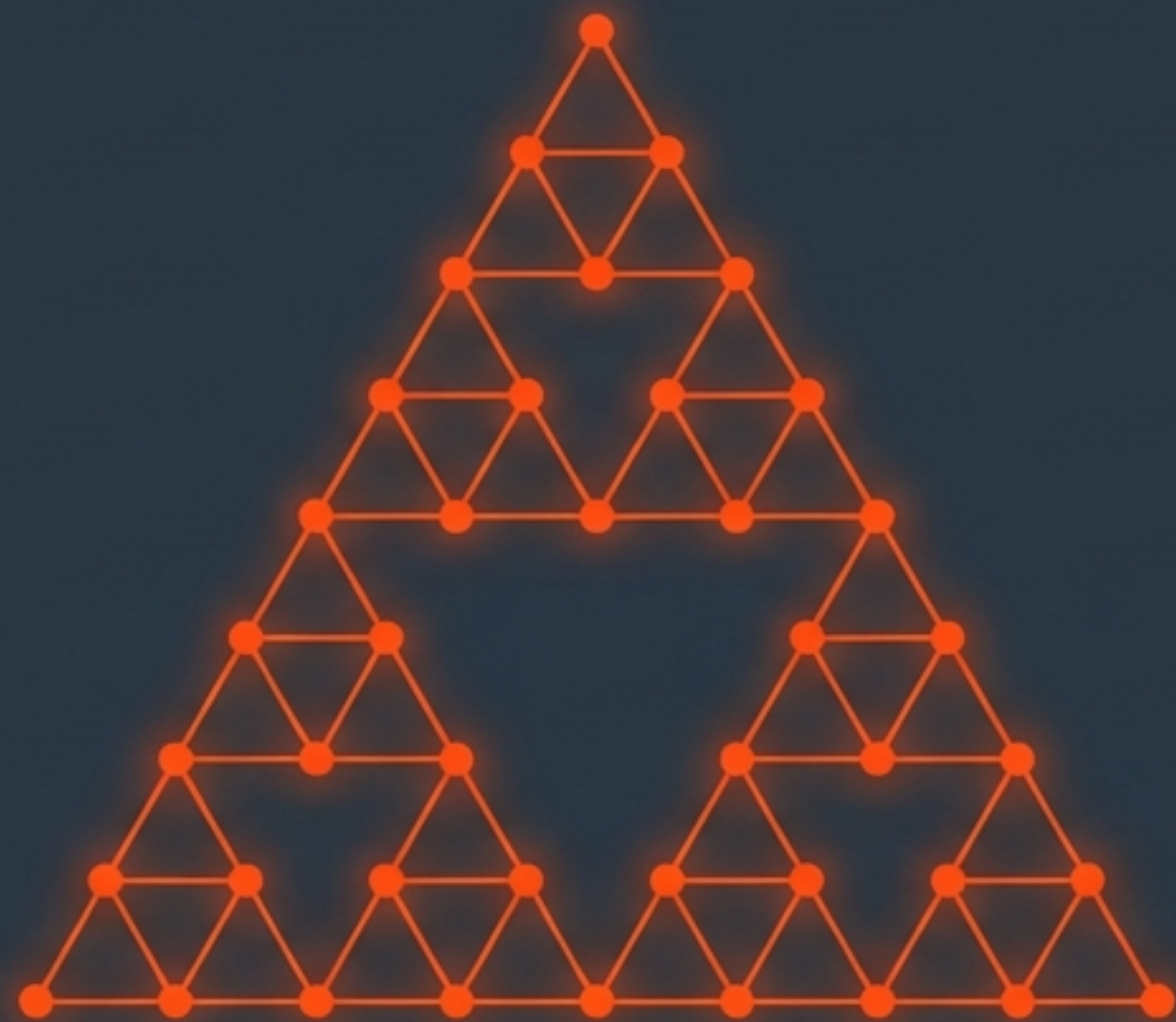


Aurphyx: Geometric Shortcuts to Fault Tolerance

Fractal-Enhanced Topological Quantum Computing



Principal Investigator:
Ross A. Edwards

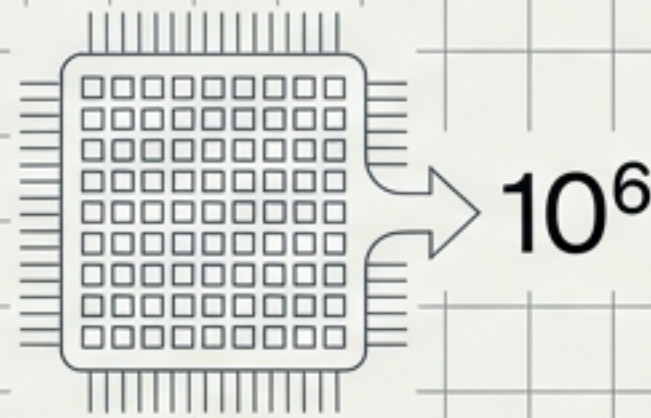
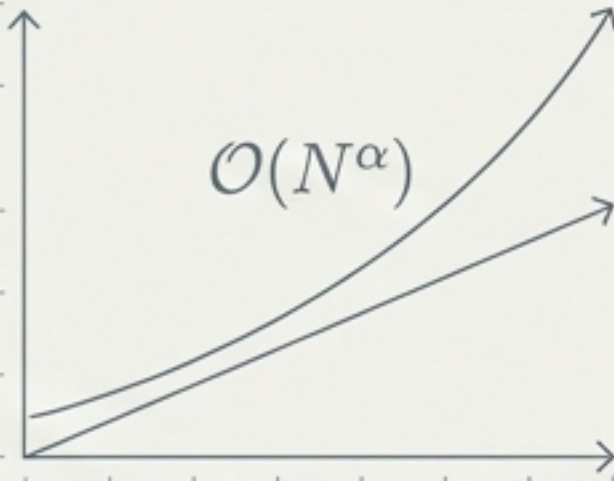
Program Target: DARPA Quantum
Benchmarking / NSF QLCI

Current TRL:
2 → Target TRL: 4

The Constraint: Standard Scaling

Polynomial error overhead.

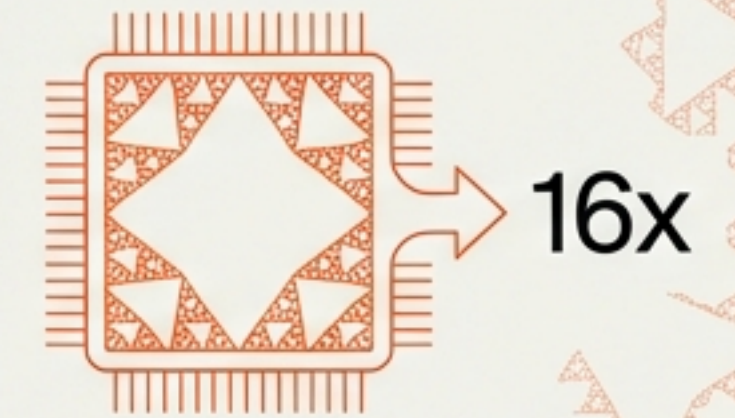
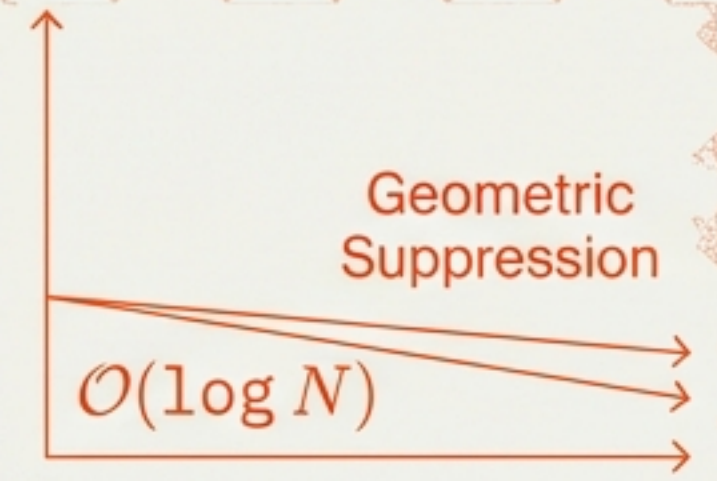
Manufacturing barrier at 10^6 physical qubits.



The Breakthrough: Fractal Scaling

Geometric error suppression.

16x overhead reduction.



$10^4 \times$

Larger accessible Hilbert space

16x

Reduction in error correction overhead

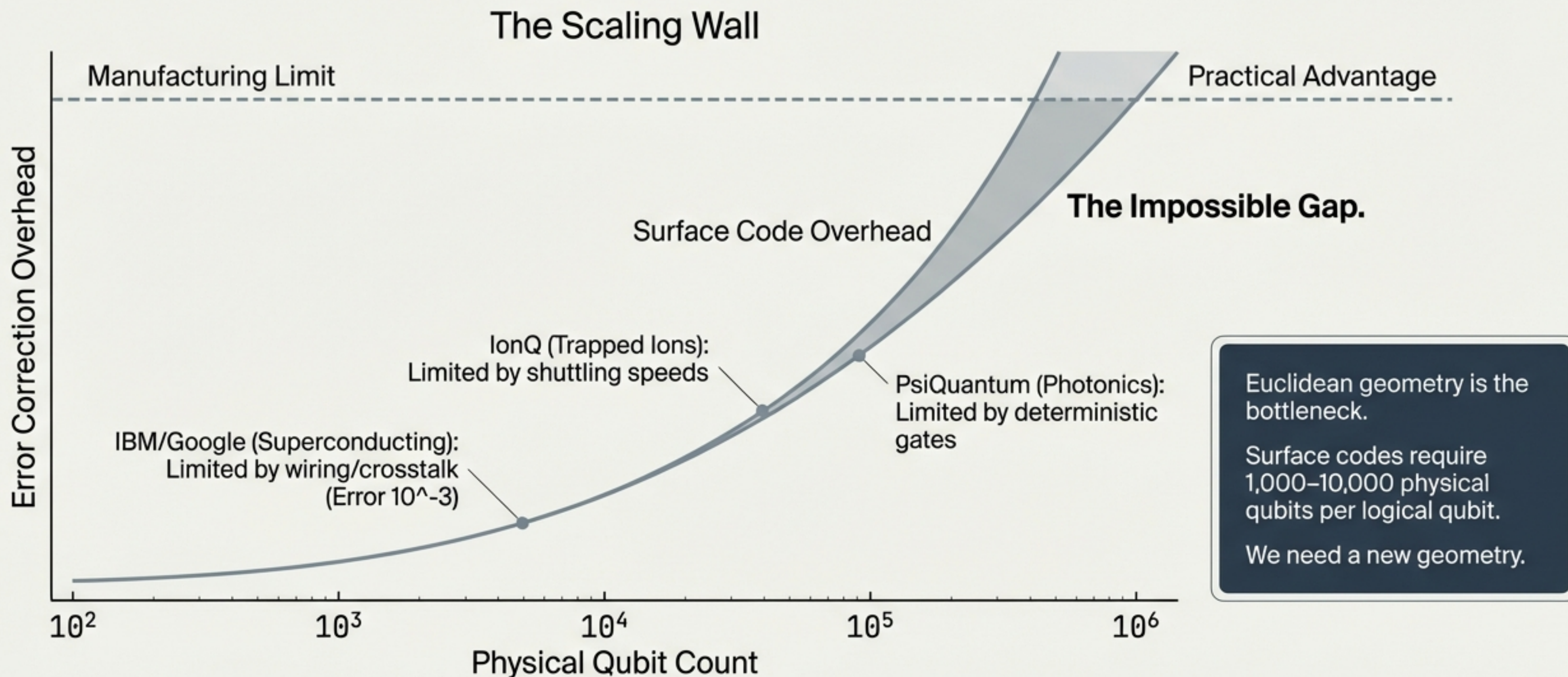
\$1.25M

Phase I Funding (18 Months)

We replace physical redundancy with geometric complexity.

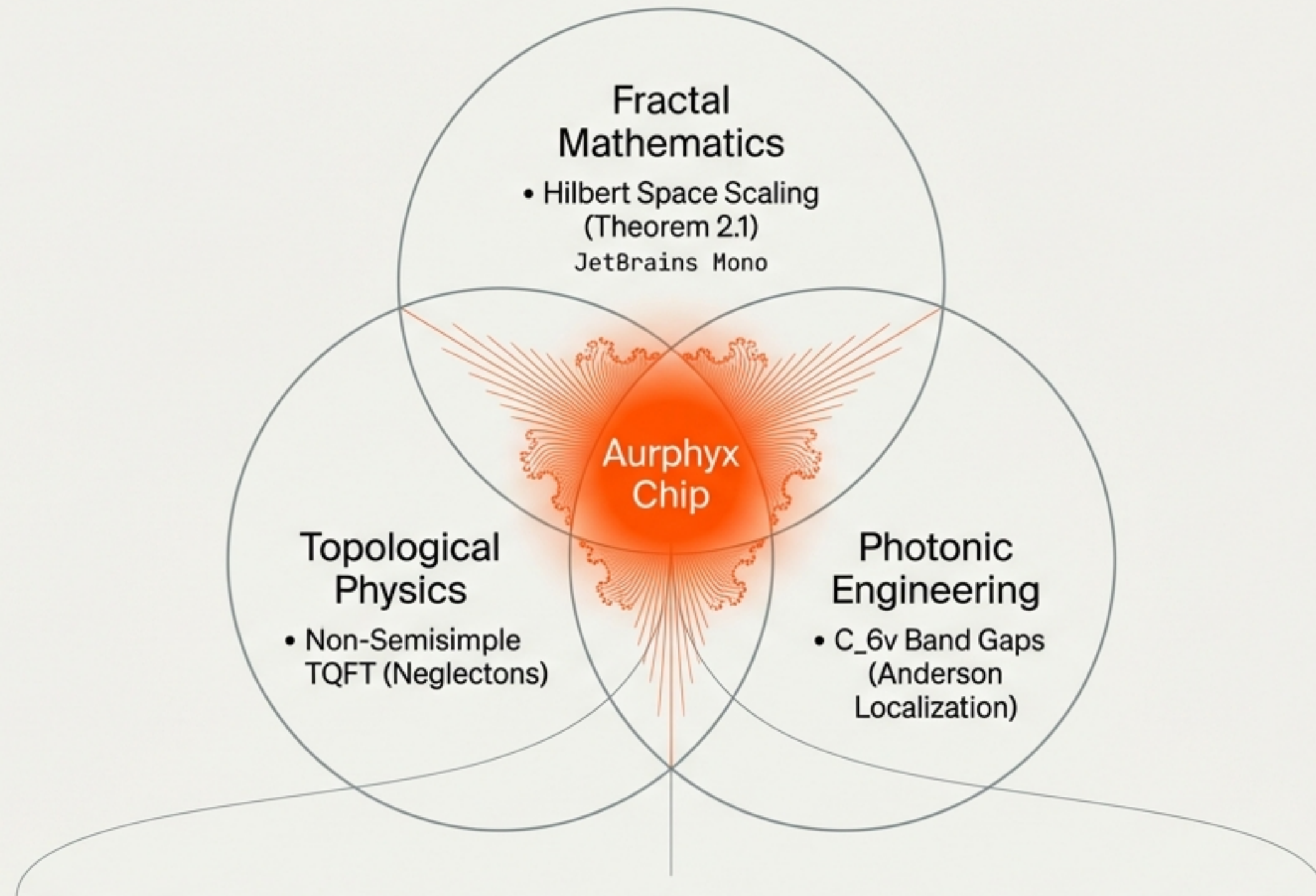
The Problem: The Quantum Scaling Crisis

Current architectures hit a manufacturing wall before reaching fault tolerance.



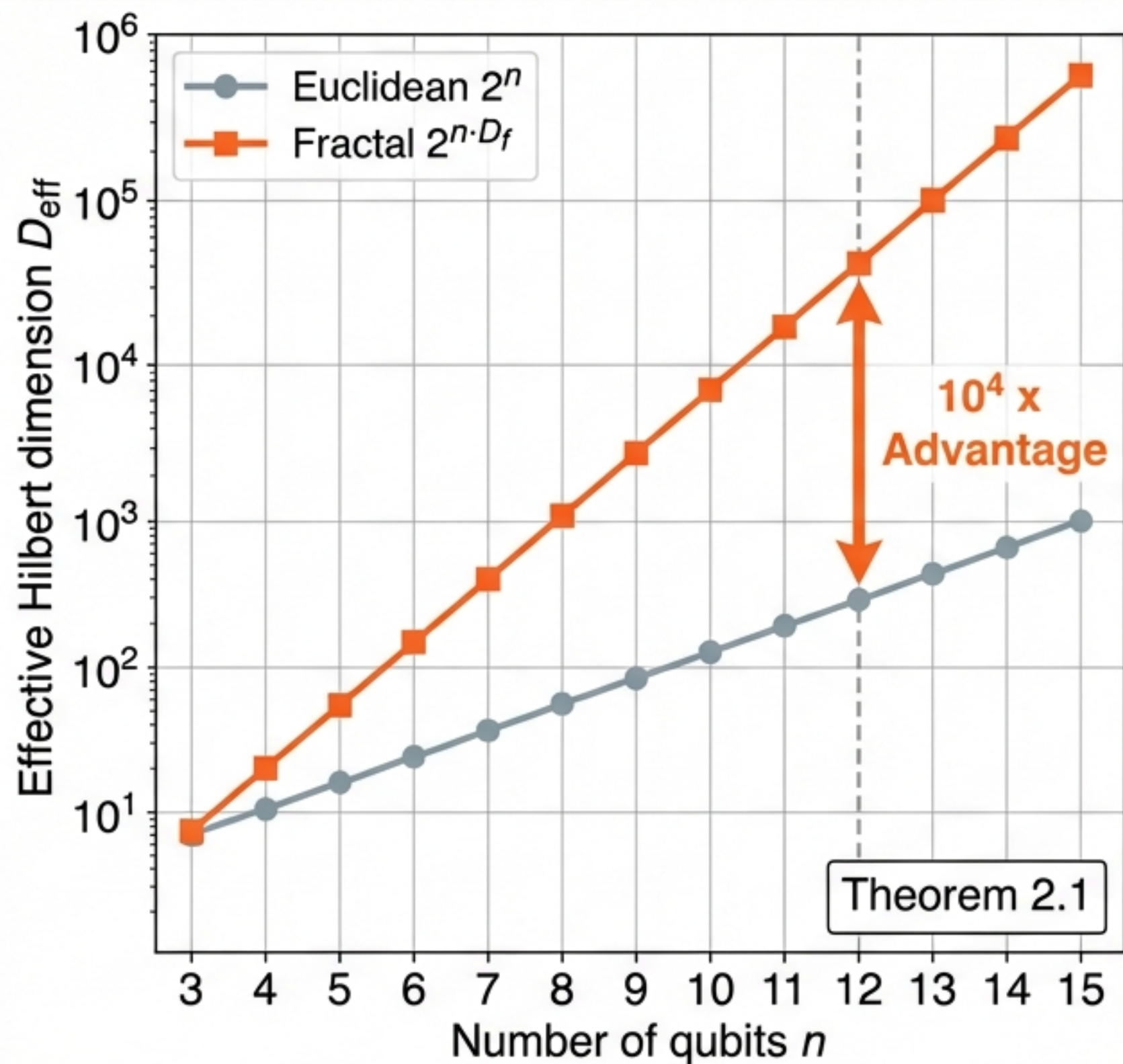
The Paradigm Shift: Aurphyx Architecture

Convergence of Three Breakthroughs.

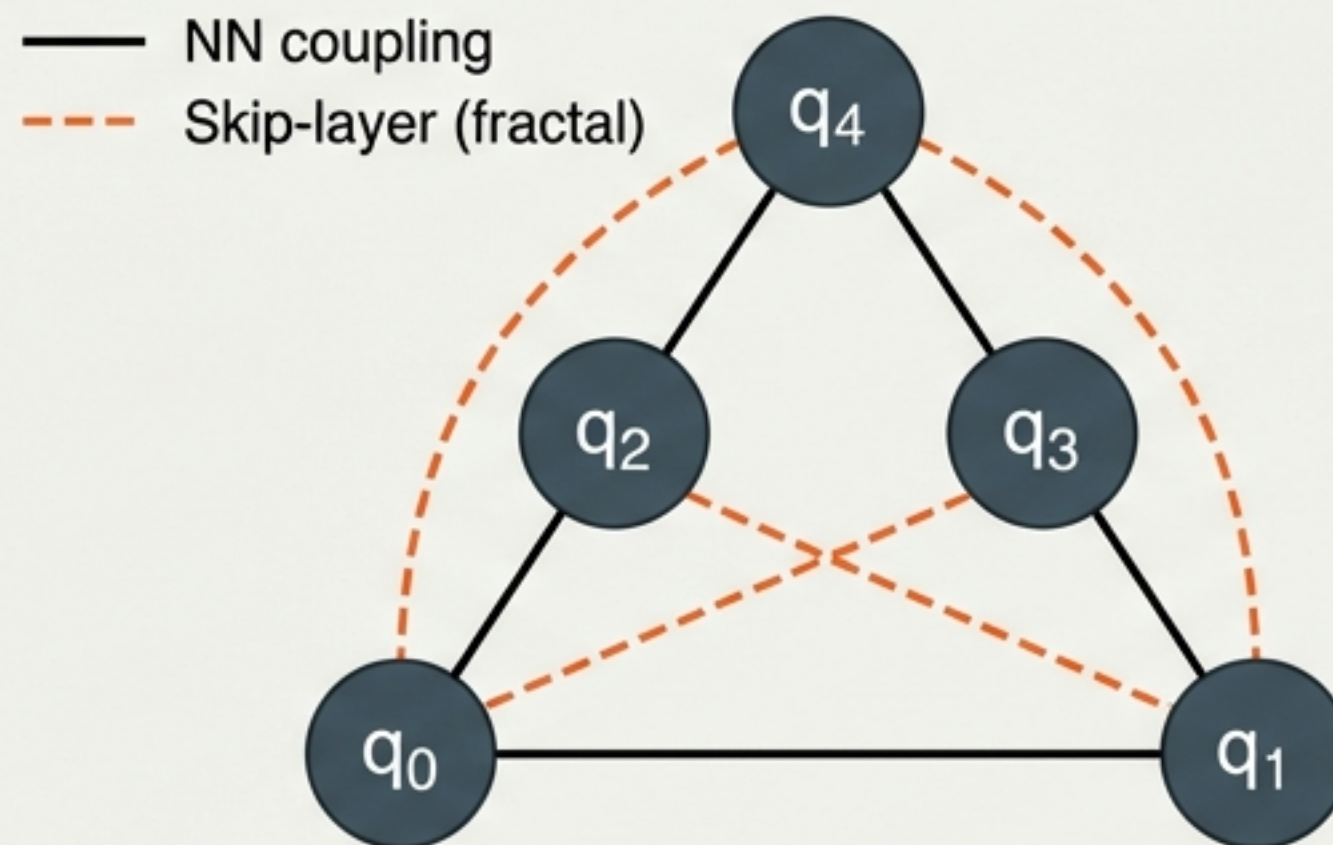


Aurphyx integrates fractal geometry constraints, zero-dimension topological particles, and photonic band engineering to suppress errors at the hardware level.

Innovation I: Fractal Hilbert Space Scaling



Fractal Connectivity Pattern

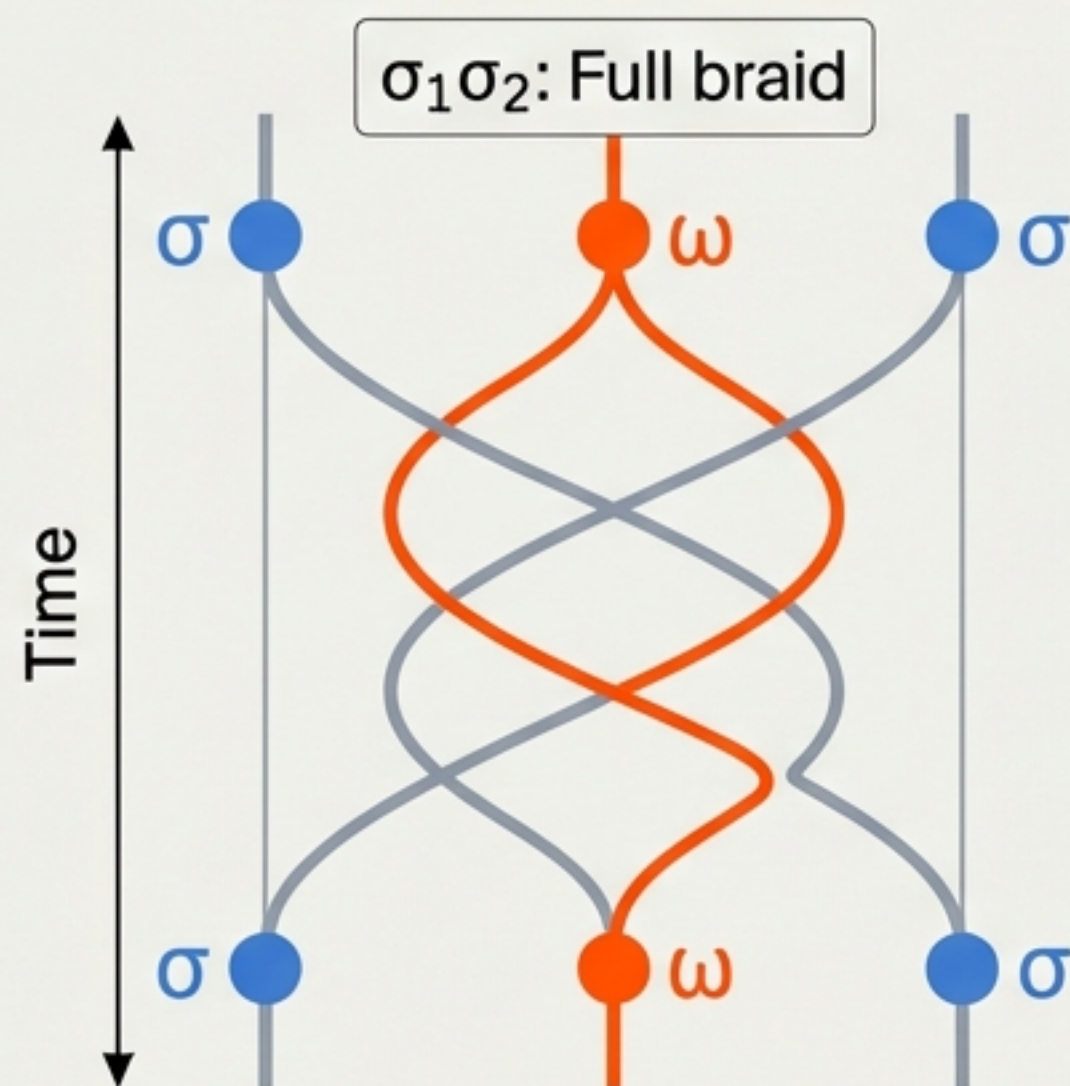


Standard grids scale linearly. Fractal lattices scale super-polynomially by using hierarchical shortcuts (Hausdorff dimension $D_f = 1.585$) to access state spaces usually requiring thousands of qubits.

$$\dim(H_{\text{acc}}) = 2^{n \cdot D_f^{\alpha(k)}}$$

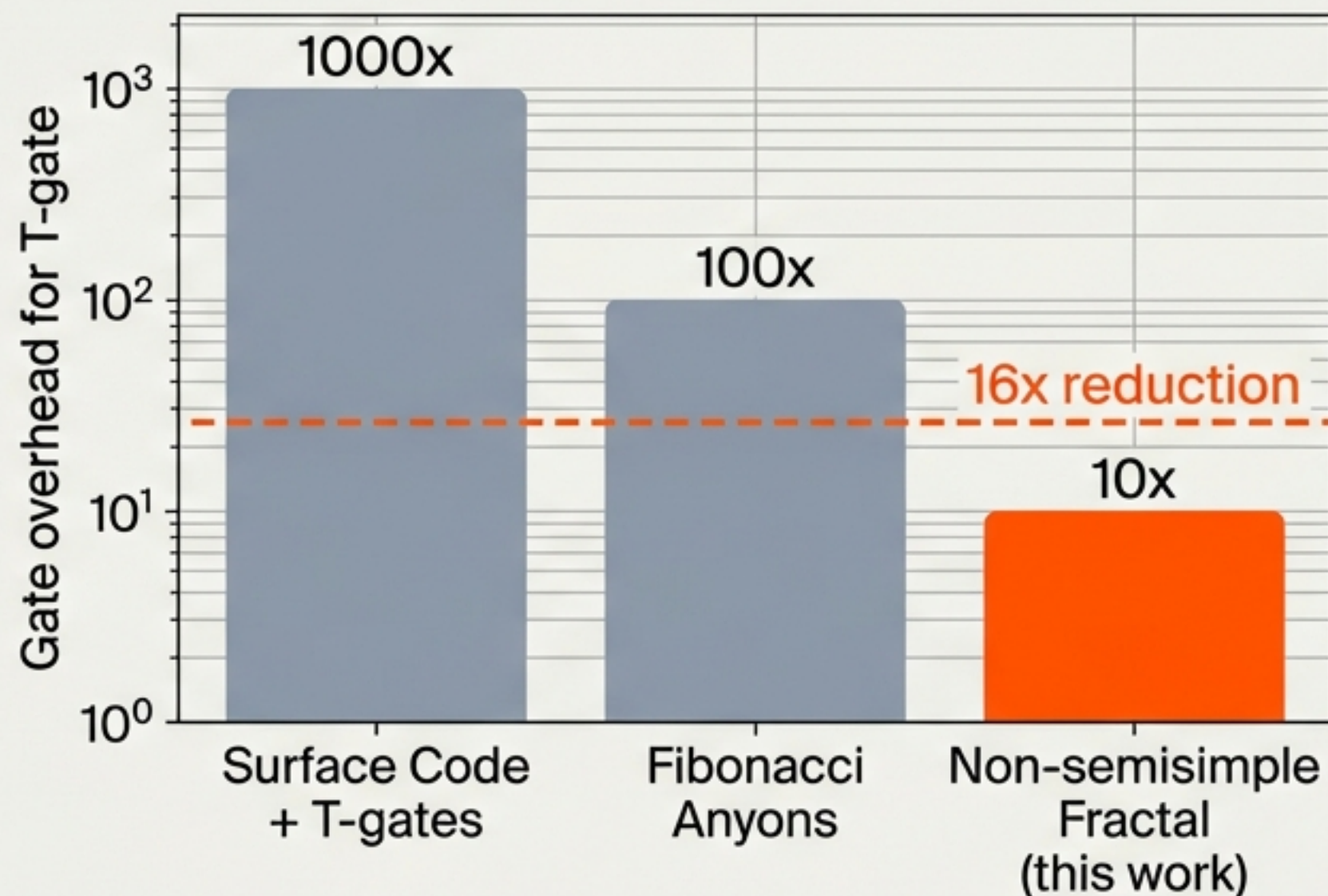
Innovation II: Non-Semisimple TQFT & Neglectons

Anyon Braiding Worldlines



The Neglecton (omega): A particle with Zero Quantum Dimension but non-trivial braiding phases.

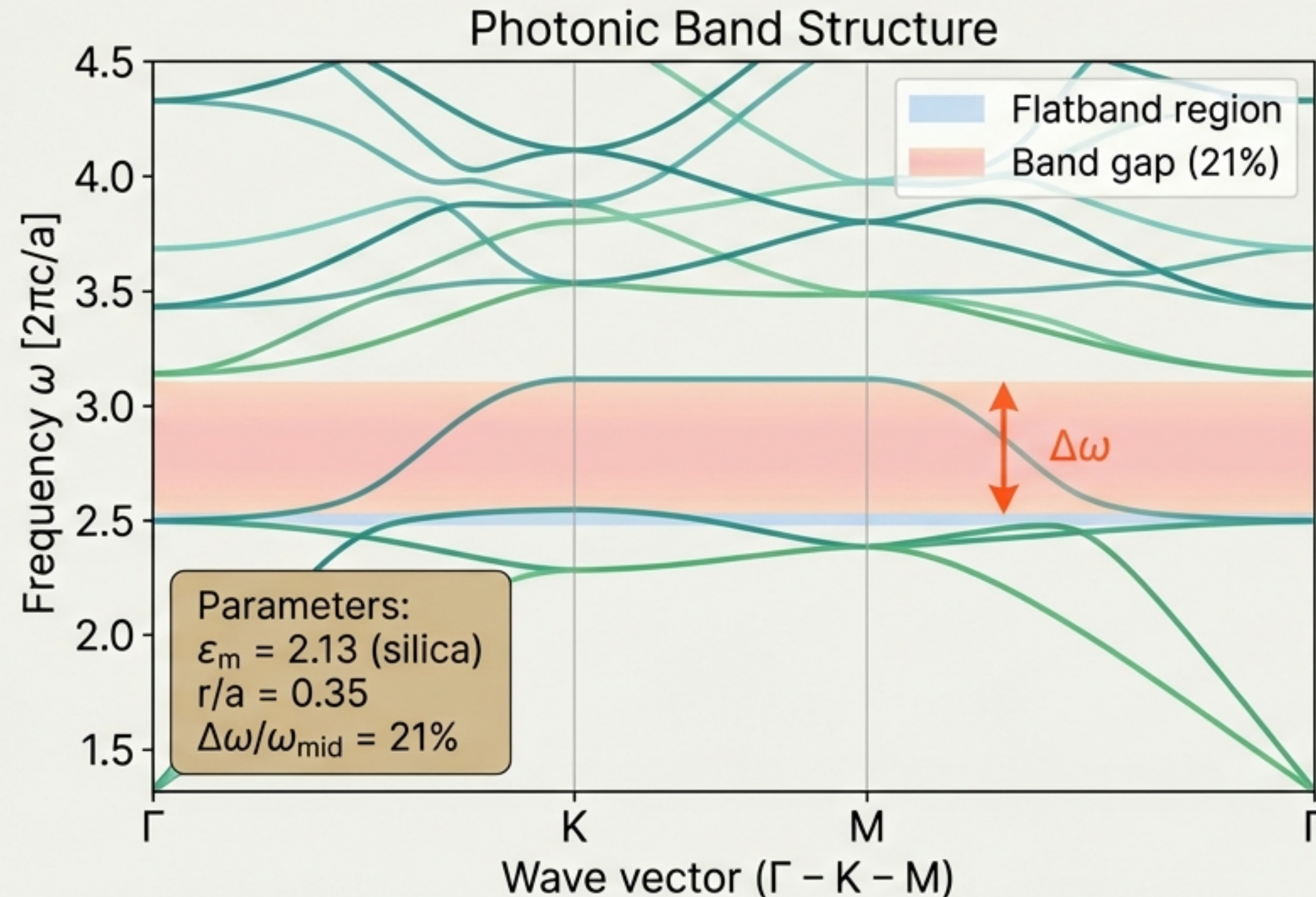
Universality Gate Overhead



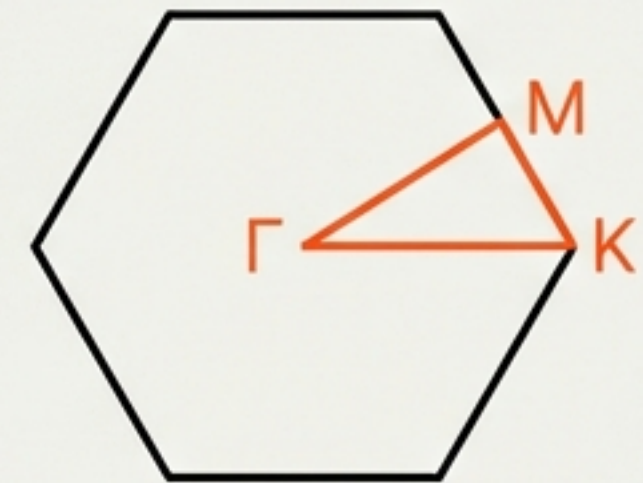
Unlike standard TQFT which requires expensive Magic State Distillation, Neglectons provide a universal gate set via braiding alone.

Innovation II: Photonic Band Engineering

Creating a physical “Quiet Zone” for Quantum Information.



Brillouin Zone

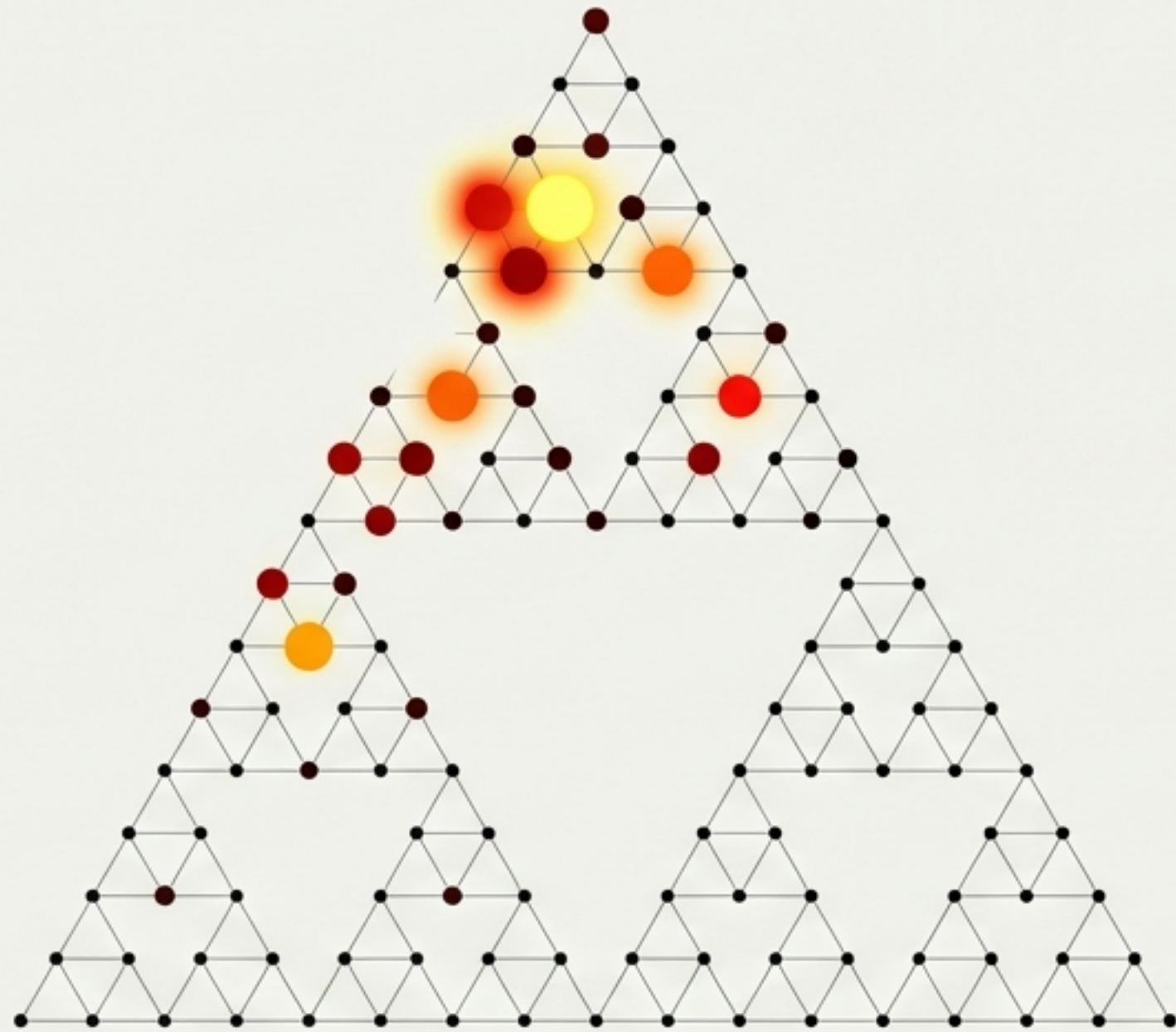


Key Metrics

- **Lattice:** Hexagonal C_{6v} symmetry
- **Band Gap:** 21% gap-midgap ratio (Telecom wavelengths)
- **Group Velocity:** $v_g < 0.01c$ (Slow light enhances interaction)

The 'Secret Sauce': Anderson Localization

Geometry itself traps errors.



Localized Wavefunction (k=4)

Mechanism: Spectral Dimension
 $d_s = 1.36 < 2$.

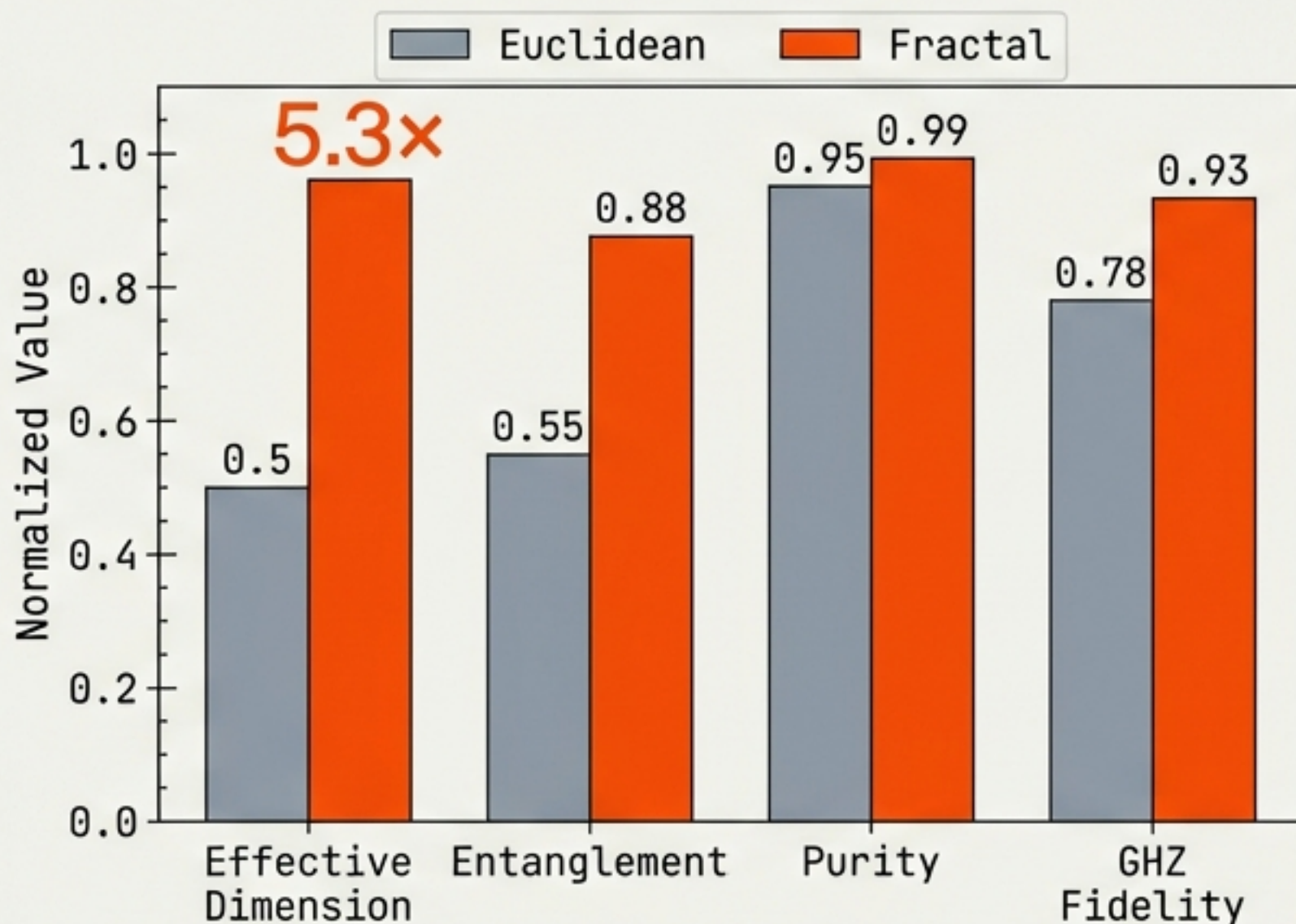
In standard lattices, errors diffuse. In fractal lattices with $d_s < 2$, errors are mathematically **forced to localize** (**Anderson Localization**) even with weak disorder.

Result: Coherence time **boosted by 16x** because errors literally cannot diffuse through the fractal structure.

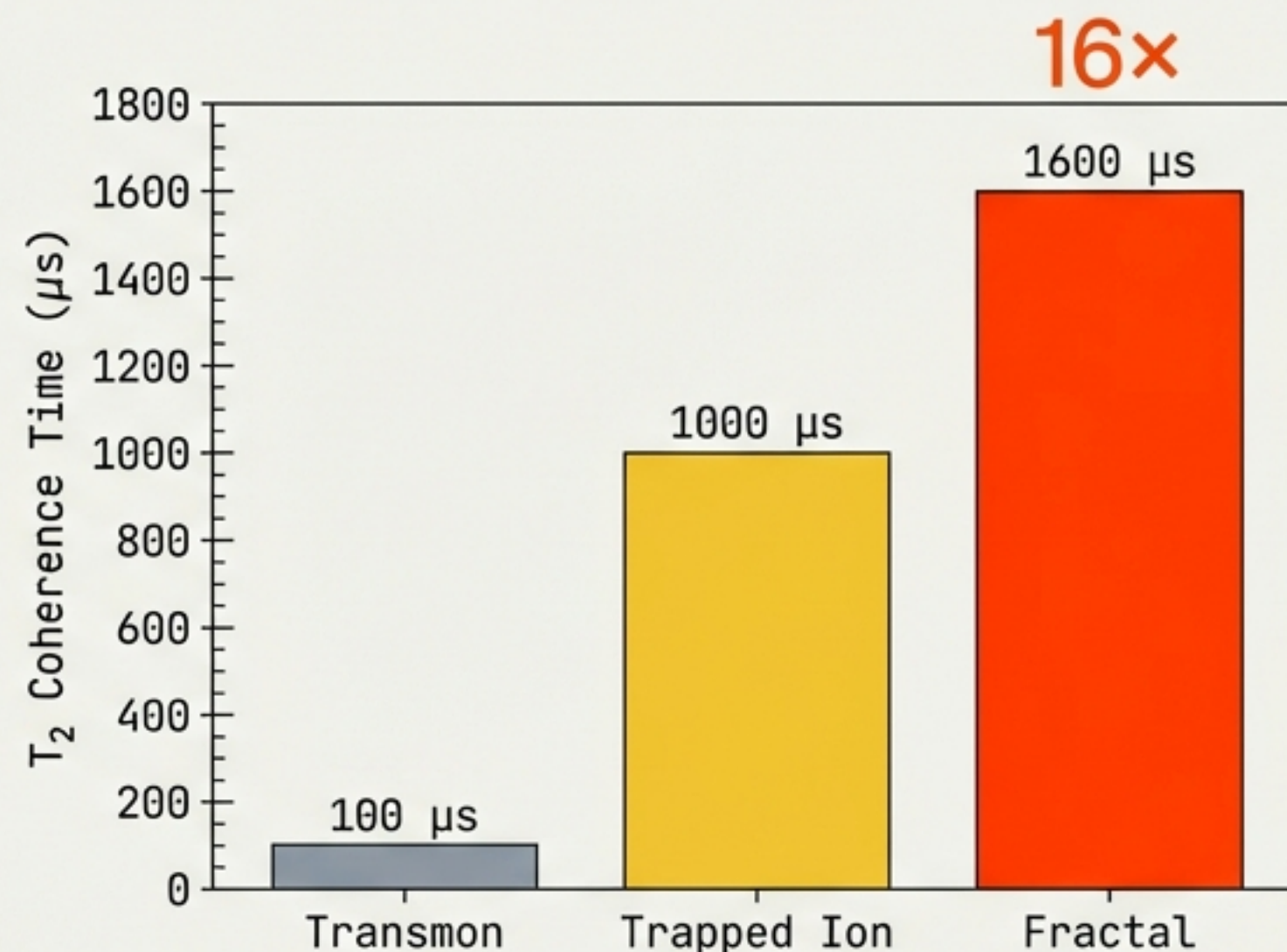
Computational Validation

Simulations confirm theoretical advantage.

Qiskit Simulation: n=5 Qubits



Coherence Time Comparison



Qiskit validation confirms 7.1x advantage at 5 qubits, consistent with theory.

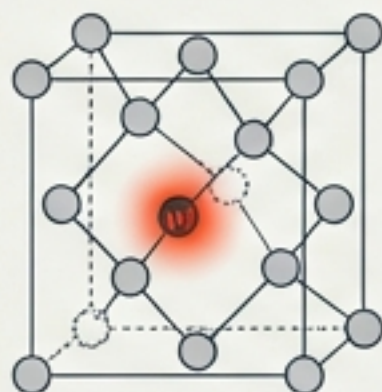
Experimental Strategy: Four Parallel Protocols

Validating the physics across complementary platforms to reach TRL-4.

P1: NV Diamond

Testing Coherence

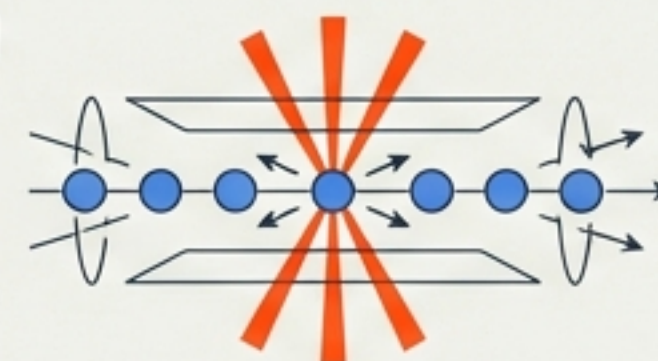
\$115K



P2: Trapped Ions

Testing Hilbert Scaling

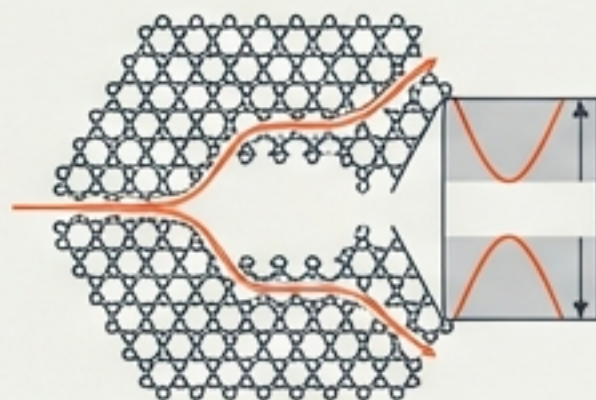
\$395K



P3: Photonics

Testing Band Gaps

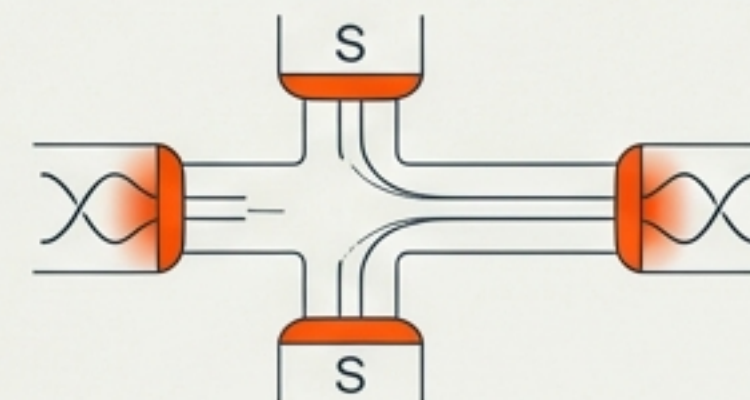
\$48K



P4: Majorana

Testing Topology

\$365K



Protocol Deep Dives: Coherence & Scaling

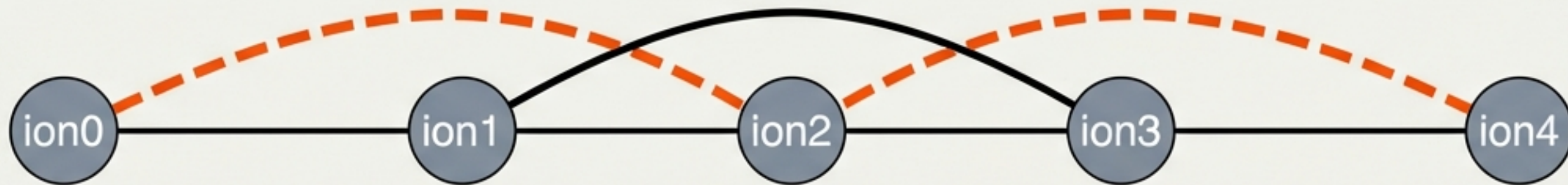
P1: NV Centers

Fabrication Process Flow



Objective: Demonstrate **>3.2x coherence improvement** on Sierpiński-patterned diamond via E-beam lithography.

P2: Trapped Ions



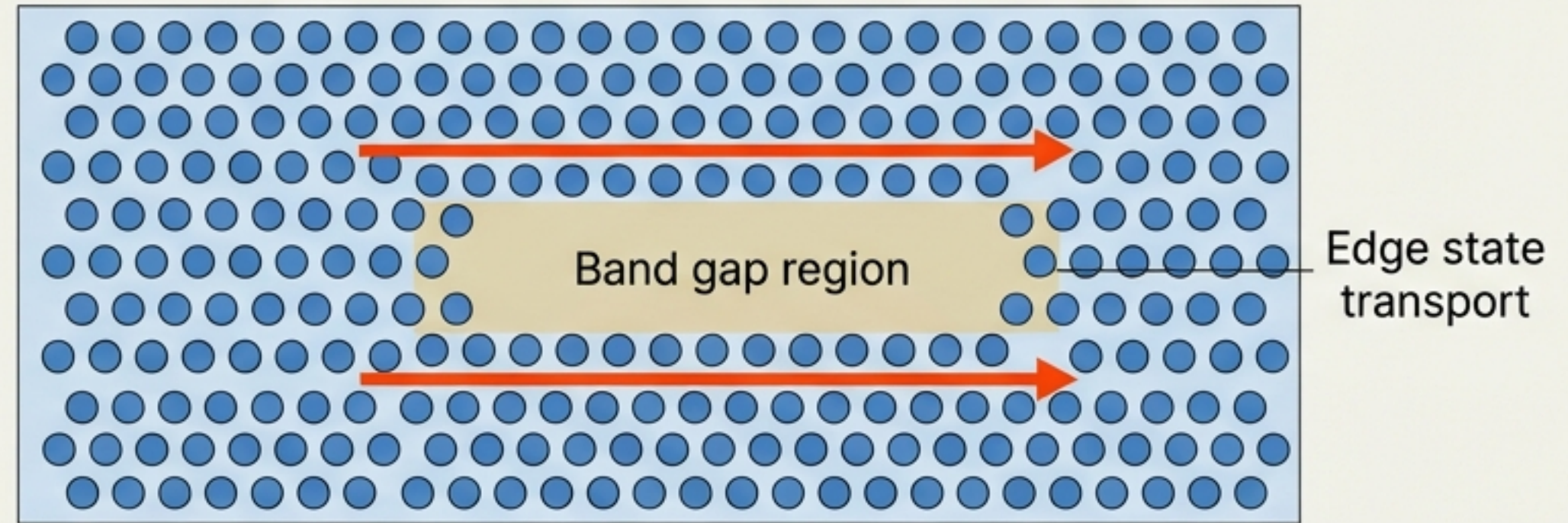
Objective: Program fractal connectivity via Mølmer-Sørensen gates on Sandia HOA traps.
Target: Measure effective dimension ratio $D_{eff} \geq 10x$.

Protocol Deep Dives: Band Gaps & Topologies

P3: Photonics

Objective: Femtosecond laser writing in fused silica to verify 21% Band Gap and edge states.

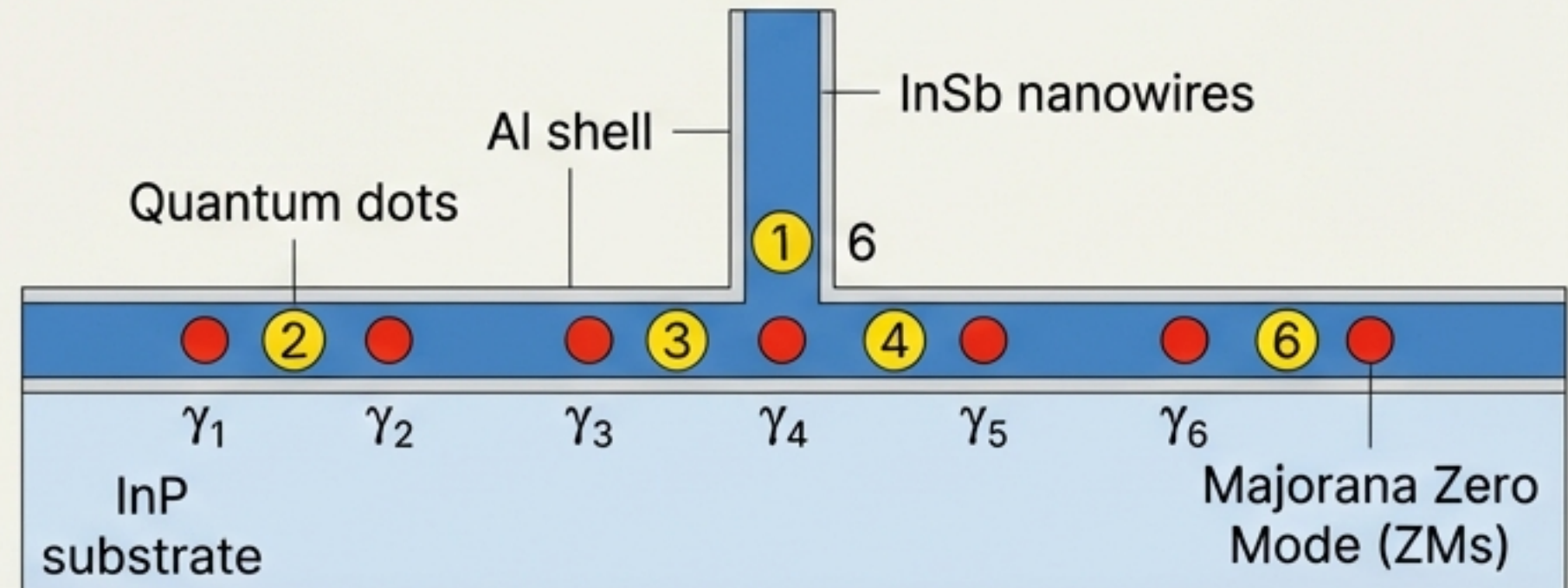
(Top View) Hexagonal Photonic Crystal



P4: Majorana

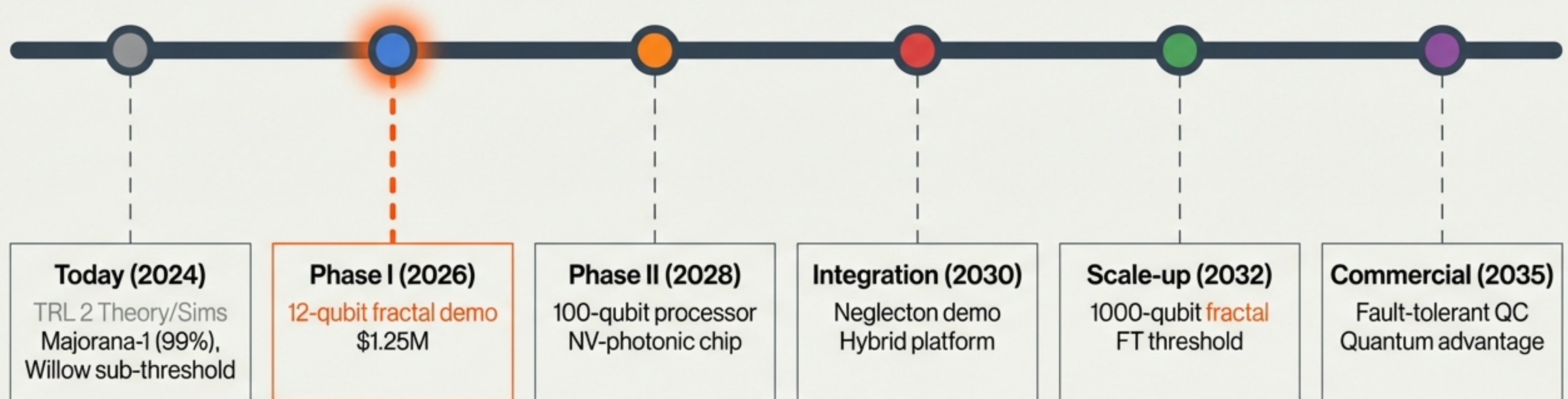
Objective: T-Junction InSb/Al nanowires to demonstrate gate fidelity $\geq 98\%$ (1.5x improvement over linear).

T-Shape 6-Dot Majorana Device



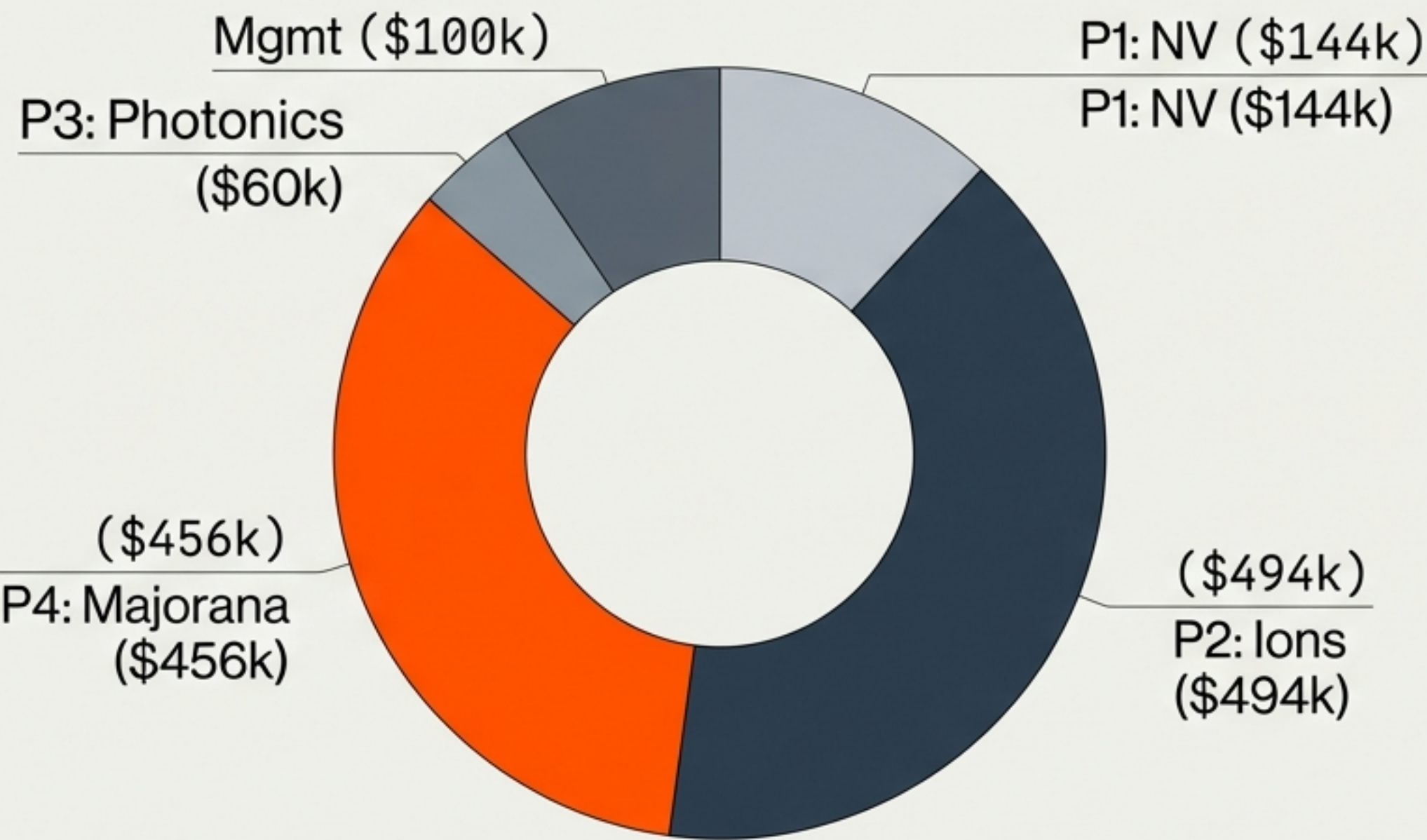
Roadmap & Milestones

18-Month Execution Plan to TRL 4



Budget & Team

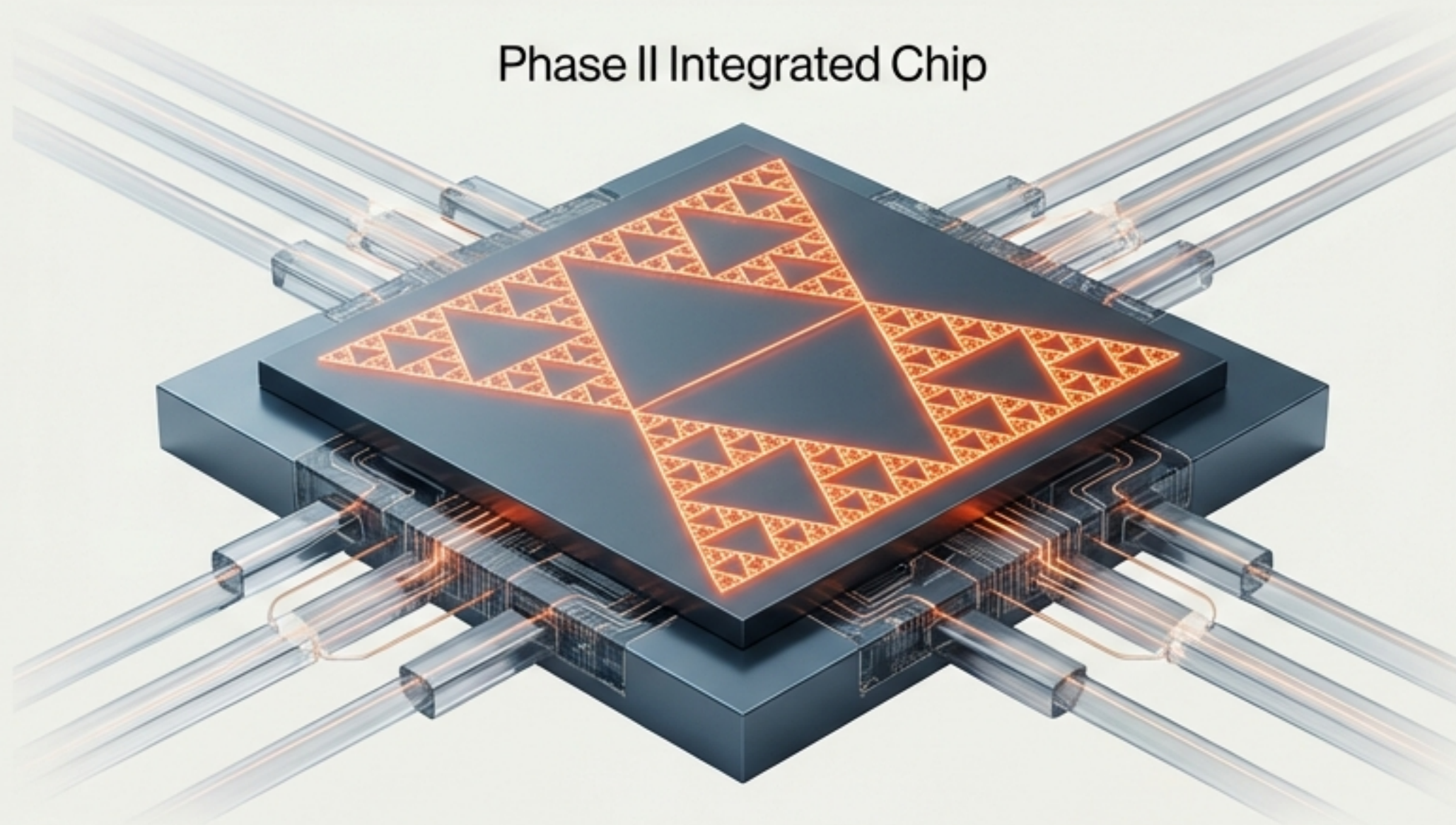
Total Ask: \$1.25M (18 Months)



Key Partners

- MIT.nano**
(Fabrication)
- Sandia National Labs**
(Ion Traps)
- Microsoft Station Q**
(Theory)
- QuTech/Delft**
(Nanowires)

Conclusion: A Geometric Shortcut



Cryptography:
Shor's algo with **10x**
fewer physical qubits.

Simulation:
Quantum chemistry with
 10^4 x state space access.

Manufacturing:
Bypassing the '**Million**
Qubit' barrier.

Aurphyx is the geometric shortcut to the Quantum Age.